

Midterm 2 Review class - November 14

Problem 1. You are selling cars and giving the following warranty. When the buyer asks you to repair the car for the first time, he will give you 1000\$. If he comes for the second time you will return him these 1000\$ and give him 1000\$ more. For every subsequent repair no money is exchanged like this. The 1000\$ amounts all go from and into the “secret account”, and you’re goal is to maximize the expected gain on this secret account for every car you sell. You know that the number of times buyer comes to repair the car is a Poisson random variable with parameter λ . What is the expected gain in the secret account for every car you sell? Which λ would maximize this value.

Problem 2. Pair of random variables (X, Y) has range

$$\{(2^{-k}, 2^{-l}) | l, k \text{ are integers and } 1 \leq k < l\}.$$

The joint probability mass function is given by

$$p_{X,Y}(2^{-k}, 2^{-l}) = C2^{-k-l},$$

for some constant C . Find the value of C . Find the marginal probability mass functions for both X and Y and evaluate $\mathbb{E}(X)$ and $\mathbb{E}(Y)$.

Problem 3. We have n guests and n different locked boxes. Assume that k -th guest has the keys to open the first k boxes. In other words, the first guest has only the key for the first box, the second guest has the keys to first two boxes, ..., the last guest has all the keys. The n boxes are distributed among the n guests uniformly at random (each guest gets exactly one box). What is the probability that the k -th guest opens his box? What is the expected number of boxes that can be opened by guests?

Problem 4. Let X be a standard Normal random variable, and Y a geometric random variable with parameter $1/4$. If X and Y are independent compute $\mathbb{E}(e^X)$, $\mathbb{E}(e^Y)$ and $\mathbb{E}(e^{X+Y})$.